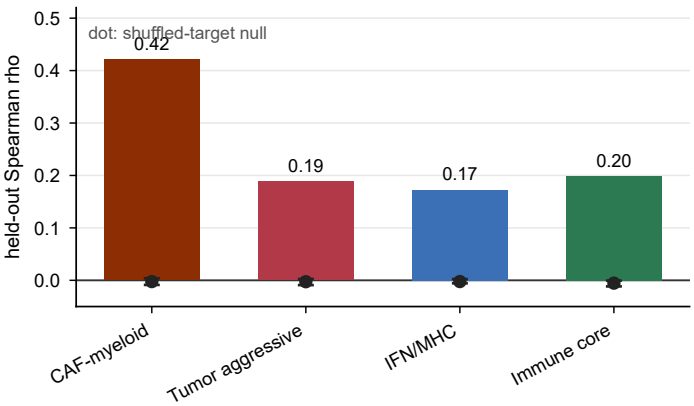
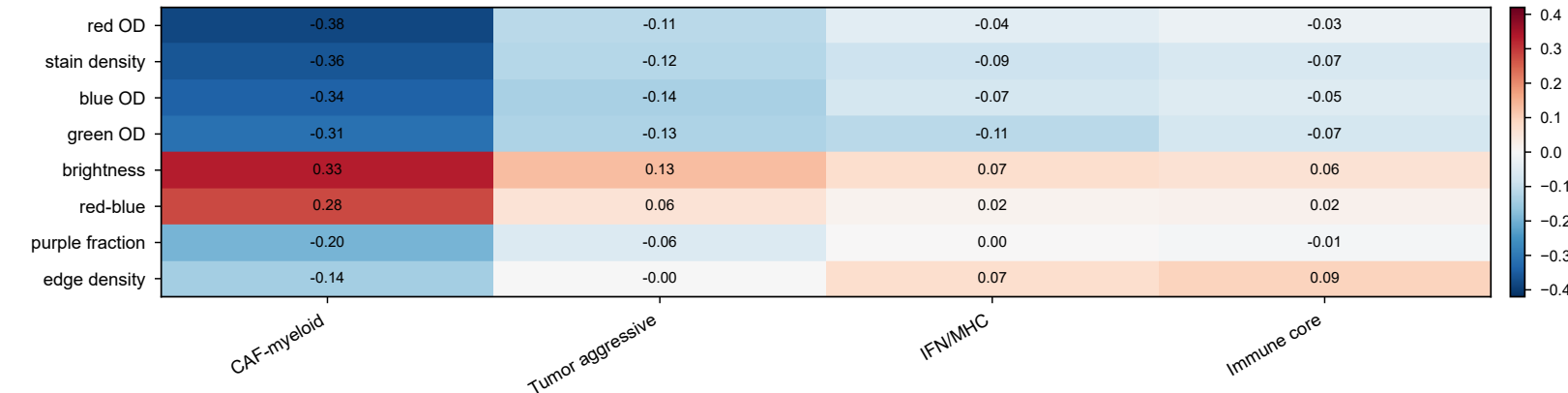


Supplementary Module 5 | Pathology bridge, TCGA context and TLS/claim boundaries

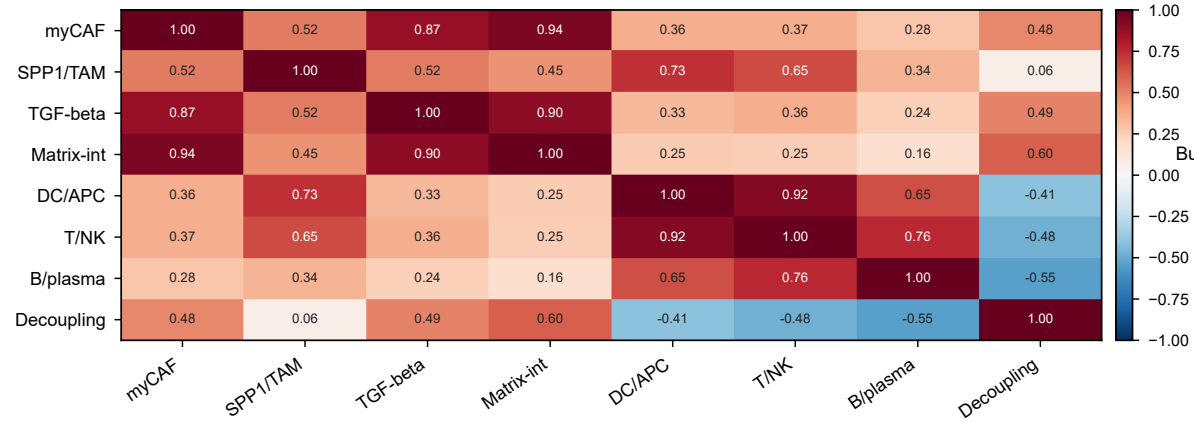
A H&E patch bridge exceeds shuffled target



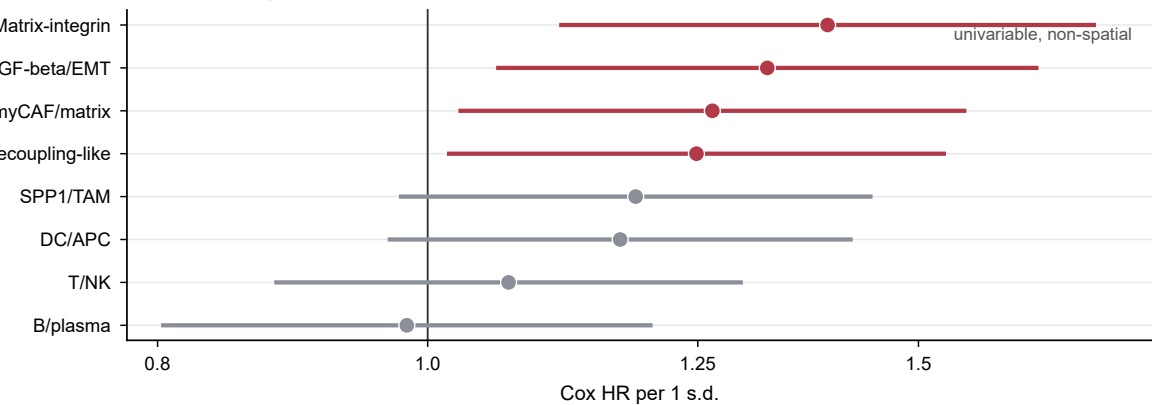
B H&E feature-direction map



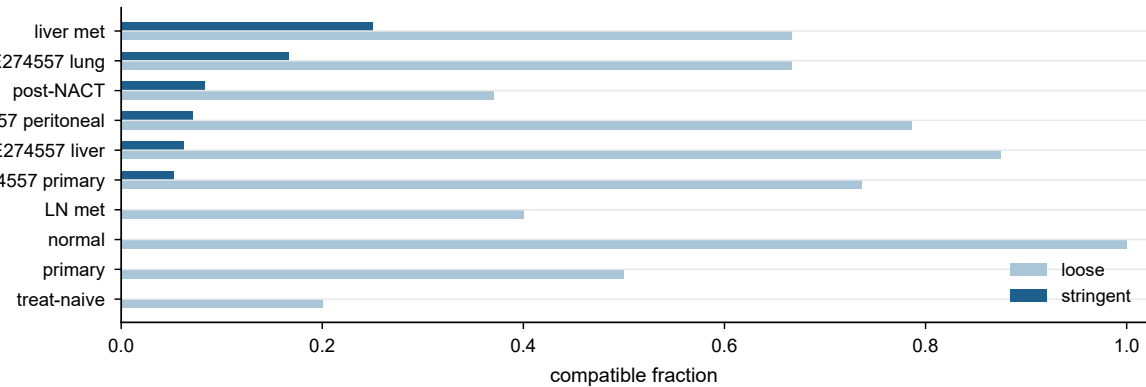
C TCGA PAAD bulk-axis context



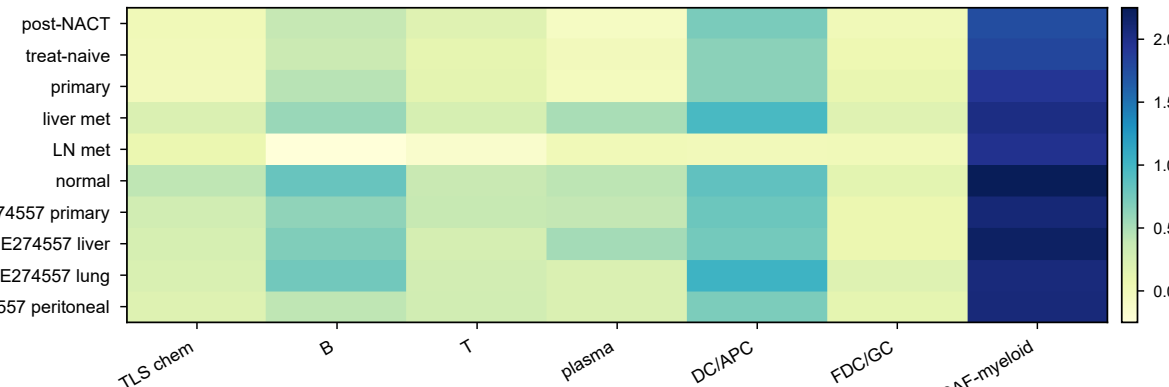
D Exploratory TCGA survival context



E TLS-maturity stress test

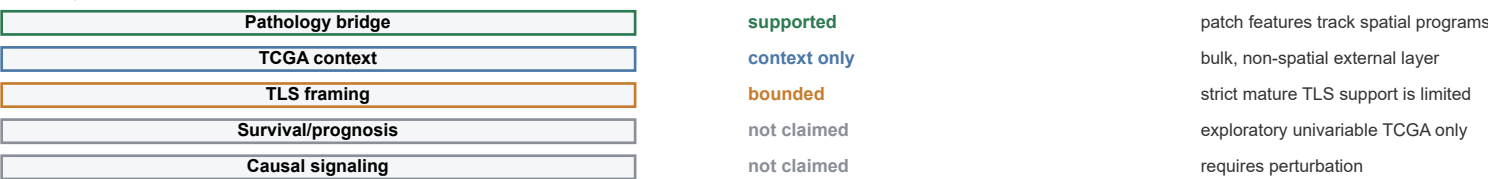


F Immune-hub markers do not equal mature TLS



G

Boundary map for reviewer-risk claims



Module role: strengthens translational plausibility while narrowing unsupported claims.